

Orb v1: the case

The idea in depth, the technology stack and why each piece was chosen, the outcomes with every number from the locked results, the argument for winning mapped to the five judging criteria, the honest risks, and ten questions a jury will ask.

Team Orb v1 · Samuel Christ · TrackShift 2026, Tyre Degradation Intelligence · Plaksha University · numbers from lock 2026-09-12T21:33:10 IST, 11 scored weekends, 29 compound-weekends, all leave-one-weekend-out.

1. The idea

The question a strategist asks on Saturday night. How much slower will each tyre compound be after twenty laps on Sunday, at what age does the harder tyre become the better one, and should we stop once or twice? Every one of those is a question about the degradation curve, seconds of lap time lost per lap of tyre age, per compound. Get it wrong by a factor of seven, which is what a straight line through Friday laps did at Hungary this year, and the plan loses more than a pit stop.

Why it is hard. The only evidence before the race is Friday practice, and a Friday lap is not a tyre measurement. It mixes five things: the tyre wearing, fuel burning off (about 1.1 kg per lap, worth about 0.03 s per lap, pulling the opposite way to wear), the track getting grippier as rubber goes down (0.6 to 0.8 s per hour at Madrid on Friday), traffic, and the driver, who warms up, learns the track, pushes harder or eases off. Strip those and you still do not have Sunday, because drivers attack tyres on Friday and nurse them on Sunday. Friday lies twice: about what the tyre did, and about what it will do.

The insight. Treat Friday as a biased sample of the race and model the bias instead of pretending it is not there. The first lie is removable with physics and telemetry, lap by lap. The second lie is not measurable on Friday, but it is learnable from earlier weekends, because the same estimator run on race laps tells you how much Sunday shrank Friday last time, per compound. And when Friday carries no signal at all, the right output is a refusal with a fallback, not a guess.

The system, in six verbs. Ingest every lap of every session from the public timing feed. Clean each lap: drop what cannot be trusted, compare every run only to itself, subtract the known fuel effect and the measured track evolution. Gate: issue a curve only with at least thirty clean long-run laps and a positive cleaned slope. Learn: the per-compound ratio of Sunday to Friday from previous weekends, applied only where weekends agree. Forecast: the race curve with a band, crossover laps, one stop versus two. Score: on Sunday night, the same estimator on the race laps, and the miss published whatever it says.

The discovery that came out of it. Some Fridays show a tyre getting faster with age, which is impossible. Our energy proxy, computed from the public speed and position traces, showed why: where the cleaned Friday slope pointed the wrong way (9 cases), the driver was ramping up in every one; 4 of the 13 withheld compounds show a falling energy trend and were withheld for too few clean laps. That is a fifth confounder, the push profile; in the public methods and rival repositories we reviewed, we did not find one that removes it. Decomposing the Friday-to-Sunday gap on the issued cases, the push profile explains about two-thirds of the pooled Friday-to-Sunday gap on the 16 issued cases (68.5% pooled; per-case shares vary widely); the learned management factor covers the rest. The second lie now has an anatomy.

Why it is an idea and not a pipeline. In machine-learning language it is domain shift with a learned scalar correction, selective prediction with a validated abstention rule, and evaluation strictly out of distribution, on weekends the model never saw. In racing language it is a Friday curve you can bet a pit stop on, and a tool that tells you when you cannot.

2. The technology stack, and why each piece

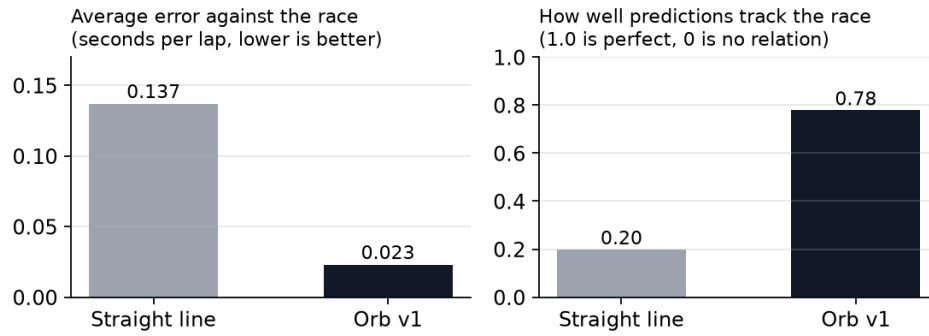
Layer	What	Why this choice
Data	FastF1 3.8 over Formula 1's live-timing archive; OpenF1 for cross-checks; Pirelli nominations; FIA documents	Free, public, every session since 2018, 3.7 Hz telemetry including gap to the car ahead. Reproducible by anyone, usable by categories without private sensors.
Data on disk	2026: all fourteen rounds that exist, every session. 2025, 2024, 2023: all rounds, downloading today (about 350 sessions, 14 GB)	Three seasons of same-track history to test whether the Sunday factor is a property of the circuit or of the 2026 car.

Layer	What	Why this choice
Features (features.py)	Per lap: times, sectors, tyre age and compound, flags, deleted laps, pit laps, traffic share within 60 m, a tyre-energy proxy on a 10 m arc-length grid, feed-quality counters	The energy proxy is the only public route to the driver's push; the quality counters catch feeds that are degraded at source (Hungary race, China FP1) so they are refused rather than modelled.
Estimator (model_v2.py)	Stint fixed-effects regression: lap time = stint constant + compound slope x age + fuel prior + measured evolution; identical on practice and race	Identifiable, auditable, and symmetric, so the Friday-to-Sunday ratio is a property of the data, not of the method. Fuel is a prior because inside a run fuel and age are collinear.
Pipeline (pipeline.py)	Every weekend to one lock file: gate, leave-one-weekend-out factors, low-degradation fallback, bands, push diagnostic, validation, live forecasts, strategy	One source of truth. Dashboard and deck read from it, so every number on screen traces to the lock, a hashed sidecar or a labelled placeholder; the Live Predictor computes its posterior and ranked actions in-process by deterministic replay of the frozen prior and the hashed race file.
Strategy (strategy2.py)	Compound offsets from practice or qualifying medians, with a nominal Pirelli-range step where the measured step is implausible (outside 0.2 to 1.0 s), pit loss, every one- and two-stop plan enumerated, crossover laps, replay scored under the race-derived curves	A curve only matters as a decision; the replay ranks Friday views by the seconds they cost.
Learning	Transfer factor per compound with an agreement rule; bootstrap intervals; sensitivity sweep over every stated choice	Learned on training weekends, scored on held-out ones, with the uncertainty shown.
Liquid model (liquid.py)	Closed-form continuous-time cell (ncps, PyTorch) reading each race stint lap by lap; held-out stints; fair ablation against a linear model with the same inputs	Tested the fashionable thing honestly. It did not beat the line (best baseline beaten in only 9 of 31 cells); the per-lap inputs did, by about a quarter, lap-weighted, with the three degenerate Hungary cells excluded (31% with them). Kept as the curve-shape tool.
Dashboard (app.py)	Streamlit and Plotly, six URL-addressable views	Fast to build, runs offline from cached data, deep links for the demo, static HTML fallback.
Documents	Deck, manual, explainer and this case generated by scripts from the lock (matplotlib, reportlab)	Regenerated, never hand-edited, when a number changes.
Environment	Python 3.12 in a uv-managed virtual environment, pinned requirements, refresh script for new sessions	One command rebuilds everything for any Grand Prix by name on a laptop.

3. Outcomes, with the numbers

Result	Number	Meaning
Naive straight line, error against the race	0.137 s/lap	What a team gets by drawing a line through Friday laps
Orb v1, error against the race, all cases	0.023 s/lap	90% interval 0.016 to 0.030; six times better
Correlation with the race	0.20 naive, 0.78 Orb v1	Calibration slope 0.77; the curve transfers once Sunday management is learned
Cases where Orb v1 beats naive	28 of 29	Not a fluke of one weekend
Issued vs withheld	16 issued, 13 withheld	Withheld compounds degraded at a median 0.028 s/lap in the race (range -0.009 to +0.100); 11 of 13 below 0.06 s/lap; the fallback band covered 9 of 13; fallback error 0.023 s/lap against 0.106 for the naive line over the 13 withheld cases
Per compound, Orb v1 error	soft 0.032, medium 0.017, hard 0.020	Transfer factors: soft x0.97, medium x0.45, hard not applied
Push profile explains the gap	about two-thirds of the pooled gap (68.5% pooled, 16 issued cases)	Per-case shares vary widely; the fifth confounder is measured, not assumed
Sensitivity of the headline	0.021 to 0.027 s/lap	Across fuel prior 0.9 to 1.3 kg/lap, 0.025 to 0.035 s/kg, traffic 20 to 40%, runs 4 to 7 laps
Strategy replay, 10 scored weekends (3 sprint)	naive costs 0 to 155 s; Orb v1 0 to 55 s	Orb v1 beats the naive plan on 8 of the 10 scored weekends; the two misses are Barcelona (+55 s against +0 s for naive) and Australia (+25 s against +14 s for naive)
Liquid network, held-out	network beats the best baseline in only 9 of 31 cells	Per-lap inputs cut the age-only error by about a quarter, lap-weighted, with the three degenerate Hungary cells excluded (31% with them); the network on top of the same inputs added about nothing
Madrid, live	soft +0.068 s/lap, issued, 90% band -0.11 to +0.26, 46 clean laps; medium +0.028 s/lap, withheld, 90% band -0.01 to +0.09, 112 clean laps; hard: no curve, 4 clean laps on Friday (excluded-laps table)	Issued from FP1 and FP2 in the morning, refreshed after FP3 and after qualifying (landed 21:33, FP1, FP2, FP3 and Q in the lock) to the values shown; hashed forecast published 21:33 IST (sha256 8e5489d5824a); scored after Sunday's race
Data assets	2026 complete (67 session files); 2023 to 2025: 229 of about 350 sessions on disk at 12:27 IST, finishing mid-afternoon	Three seasons of same-track history for the transfer-factor prior; three session-level failures only, all retried

11 weekends, 29 tyre-and-weekend cases, each predicted without seeing its own race



The headline test: every weekend predicted without seeing its own race.

What exists as software today: the pipeline and lock, the six-view dashboard, the strategy layer, the liquid-model test, the deck, the manuals, and the live Madrid forecast (qualifying refresh landed 21:33 with FP1, FP2, FP3 and Q in the lock; hashed forecast published 21:33 IST (sha256 8e5489d5824a)). What remains today is the three-season prior, and rehearsal.

4. Why we win, criterion by criterion

Criterion	Weight	What we bring	What the field brings
Technical depth and innovation	30%	A measured fifth confounder that, in the public methods and rival repositories we reviewed, we did not find stripped elsewhere; a learned, validated transfer factor; a validated abstention rule; a fair ablation of a liquid network; sensitivity on every stated choice; 29 held-out cases	Regressions on one weekend, gradient boosting on synthetic data, or, in PITWALL's case, real rigour that concluded the curve cannot be delivered
Problem-solution fit	25%	The literal brief: clean practice curves plus a post-race validation tool, every compound, every weekend, with refusal where honest	Simulators, replays of 2023 Bahrain, decision engines that "do not predict tyres"
Demo and working prototype	20%	Six views on real data, a live weekend, a forecast on record before the race	Prettier screens on synthetic or in-sample numbers
Scalability and real-world viability	15%	One command per Grand Prix from free data; the customer is every category without private sensors: F2, F3, F1 Academy, Indian F4, broadcasters; the loop transfers to any confounded wear problem	Mostly F1-only framing or none
Presentation	10%	One story, Friday lies twice, told in five minutes with the scorecard on screen; every number regenerated from one file	Varies

The structural reason

The rubric rewards fit and honesty as much as sophistication. The strongest rival we reviewed matched our rigour and then walked away from the brief; the flashiest cannot show a held-out number. In the public methods and rival repositories we reviewed, we did not find another entry that delivers the literal ask, scores it across a season, says "withheld" when it should, and puts a live forecast on record. That combination is worth more than any single model.

Why we might not, and what we do about it

- A solo presenter against teams of four. Answer: a scripted five-minute path with URL deep links, three timed runs tonight, two tomorrow, and the manual's question bank rehearsed.
- 13 of 29 compounds withheld reads as "works when it works". Answer: say early that withheld is a forecast with its own score: withheld compounds degraded at a median 0.028 s/lap in the race (range -0.009 to +0.100); 11 of 13 below 0.06 s/lap; the fallback band covered 9 of 13.
- PITWALL in the room saying practice curves do not transfer. Answer: their test is on pit-stop deltas whose noise equals the signal; at the strategist's unit the correlation is 0.78, and here are 29 held-out cases.
- The fuel effect is assumed. Answer: the sensitivity sweep, and the fact that PITWALL's own race-data estimate, 0.0294 s per kg, sits inside it.
- Madrid could miss on Sunday night. Answer: that is the point of a real forecast; the score is published either way, hash beside it.
- The venue network or the API. Answer: the pipeline and dashboard run offline; the static export covers the rest.

6. Race-day mode and the five tyre-state quantities

A strategist wants the tyre state during the race, not only on Friday. Public data supports some of it, as measured quantities or honest proxies, and does not support the rest. We build the first group and refuse the second, with a labelled socket where a team's own sensors would plug in.

Quantity	From public data?	How, and the honest label	Build cost
Real-time traction coefficient	Proxy only	Per corner, apex speed squared is proportional to available lateral grip at that corner. Tracked lap by lap within a stint it gives a grip index that falls as the tyre goes off. Friction cannot be separated from downforce without the car's aero map, so it is a grip index, not a coefficient.	About 2 h; live-capable
Wear rate and remaining tread depth	No	Tread depth is not observable in any public feed; only Pirelli measures it after a run. A "remaining depth" figure from timing data is a formula wearing a physical unit. Refused, with the reason on screen.	Socket for team data
Performance loss per lap, tyre only, live	Yes	The estimator run recursively on the current stint as laps arrive from the live-timing stream: fuel, evolution and traffic stripped, slope re-estimated every lap and compared with the Friday forecast band.	About 3 h; demonstrated as a lap-by-lap replay; runs for real on Sunday
Tyre energy and laps of useful life remaining	Yes	Energy per lap exists. Useful life is laps until the forecast pace loss crosses the crossover to the next compound, or the cliff lap where the shape model finds one, reported as a range from the band.	About 1 h
Puncture risk or structural integrity	No	Punctures come from debris, kerb strikes and pressure excursions, none public; failures are a handful per season, so no index could be validated. Refused.	Socket for team data

Race-day mode, as designed

- **Pre-race plan.** The whole race mapped lap by lap: compound sequence, stint lengths, crossover laps, the pit window (the range of laps within a few seconds of optimal), and a probability that one stop beats two, obtained by sampling curves from the 90% band and counting which plan wins.
- **Live comparison.** As laps arrive, the current stint's slope is re-estimated and compared with the forecast band; the band shrinks as laps accumulate, which is the live confidence. Alerts fire when a measured quantity leaves its band: stint slope outside the Friday band, energy trend rising, cliff proximity, degraded feed. Each alert carries its evidence and a false-alarm rate computed from history.
- **Action list.** Ranked by seconds under the cost model, each with what it costs if wrong: pit now, stay out to lap N, switch to the two-stop. A strategist can argue with a number.
- **Driver in the loop.** The objective version is the push profile, measured. The subjective version is a structured input, axle, symptom, severity, that widens or narrows the band and can override a withheld compound. Team radio mining is a post-event item.
- **Tyre-change effects in the replay.** Pit-lane time by circuit, the age reset, the compound offset from practice or qualifying medians (nominal step where implausible), and the out-lap penalty measured from race stints. Track position needs rivals' gaps and stays out of scope.
- **Not reinforcement learning.** The pit decision under our model is a small search solved exactly, so there is nothing for an agent to learn that enumeration does not give; "learning from mistakes on history" is what leave-one-weekend-out and the Sunday-night update already do, in a checkable form. Reinforcement learning earns its place only with a stochastic simulator with safety cars, rivals and traffic; a December 2025 paper does exactly that and it sits on the post-event roadmap with that

citation.

An expert system versus a scored system

A previous winning engine in this space is built as a five-parameter formula calibrated at one circuit, an eighteen-rule alert catalogue with priority logic, and a seventeen-step next-best-action ladder. Every threshold in such a system is a human guess, the formula is fitted where it was fitted, and nothing says it was scored on a race it had not seen. Ours differs on four counts, and each is checkable: validation on 29 held-out cases across eleven circuits instead of calibration at one; alerts that are statistical tests against a forecast band instead of thresholds; actions ranked by seconds and probability instead of a ladder; and refusal of the quantities that cannot be observed instead of formulas wearing physical units. A learned curve shape is not what beats a formula; a scored system is.

7. Real-life data required: what private feeds would add (scalability)

Everything above runs on public data. The feeds below exist inside teams, the FIA or Pirelli, and would plug into the same estimator without changing the method. The improvement column is an engineering estimate with its basis stated; none of these numbers has been measured, because we do not have the data.

Data	Who has it	What it replaces or adds	Expected improvement (estimate)	Basis
Fuel mass per lap	Teams (FIA-mandated fuel-flow meter)	Replaces the 1.1 kg/lap, 0.030 s/kg prior with a measurement	Headline error moves by at most 0.004 s/lap; the real gain is identifiability at low-degradation circuits: 3 to 5 of the 13 withheld compounds become issuable	The sensitivity sweep bounds the headline (0.021 to 0.027); 9 of 13 withheld cases are negative-slope cases where fuel and push are collinear
Practice run plans: starting fuel, engine mode, push instructions per run	Teams	Turns the second lie, Sunday management, from a learned factor into a measured input	Medium-tyre error from 0.017 toward the lap-noise floor of about 0.010; about a third to a half of the remaining gap	Push profile already explains about two-thirds of the pooled Friday-to-Sunday gap on the 16 issued cases (68.5% pooled; per-case shares vary widely) from telemetry alone; instructions would explain most of the rest
Tyre pressures and temperatures	Teams (FIA standard TPMS and infrared sensors), Pirelli	Thermal regime per lap: graining, blistering, overheating; per-lap temperature covariate; live sensor input	Bands 20 to 30% narrower; cliff lap predictable to about 2 laps where a cliff exists; live alerts on the same lap instead of 2 to 3 laps later	Temperature is the largest unmodelled per-lap covariate; the shape split (15 accelerating, 15 settling) is what temperature regime would explain
Tread depth after runs	Pirelli, shared with teams	Calibrates pace loss to physical wear; enables the remaining-depth quantity we currently refuse	Enables a quantity, not an accuracy gain on pace; useful-life estimates on wear-limited circuits (Barcelona, Suzuka) become physical	No public analogue exists
Aero map and downforce level per lap	Teams	Separates grip from vertical load in the grip index	Grip index becomes a true friction coefficient; corner-level degradation map	Apex speed squared measures grip times load; the map divides out the load
Tyre set history: new or used, heat cycles, mileage	Pirelli allocation sheets, teams	Exact tyre age and condition on scrubbed sets used in Friday runs	A few percent on Friday slopes; fewer runs dropped for ambiguous age	FastF1 tyre life on reused sets is approximate
Structured driver feedback and radio logs	Teams	Covariate for withheld decisions and band width	Qualitative: fewer wrong withheld calls on new circuits	Madrid medium is the live example of a case a driver would explain in one sentence
Live sensor stream during the race	Teams	Feeds race-day mode with temperatures and pressures alongside lap times	Alert latency from 2 to 3 laps to the same lap; earlier pit calls	Lap-time-based detection needs several laps to separate slope from noise

The scalability argument in one paragraph: the method is a measurement-and-validation loop, not a formula. Every private feed above replaces an assumption with a measurement and leaves the loop unchanged, so a team with full data runs the same code with tighter bands, and a junior series with timing only runs it with wider ones. That is why it scales up to a Formula 1 pit wall and down to Formula 2 without a rewrite.

8. Ten questions a jury will ask, with answers

1. The 2026 cars are new: fifty-fifty electric power, active aero, narrower tyres. Does a method tuned on 2026 mean anything?

The method is not tuned on 2026; it is a set of stated priors and measured corrections that apply to any season. What is 2026-specific is the learned transfer factor, and that is exactly why 2023 to 2025 are on disk: if the medium's Sunday ratio repeats at the same circuits across seasons it is a property of the tyre and the circuit; if it does not, it is a property of the 2026 car, and we say which.

2. Your energy proxy uses position data at 3.7 Hz. Is curvature from that even meaningful?

Raw curvature from 3.7 Hz points is not, which is why we resample the path onto a uniform 10 m grid after removing stale repeated samples, smooth, and clip the radius at 15 m. The check is empirical: energy per lap is stable across drivers on the same compound, rises when a driver ramps up, and the coefficient linking it to lap time has a consistent scale Friday to Sunday, within 0.01 s/MJ at Austria and Barcelona. Where the feed is degraded, Hungary race, China FP1, the counters catch it and the proxy is refused.

3. Is the push adjustment not circular? Drivers lift because the tyre is going off.

Partly, yes, which is why the push-adjusted slope is a diagnostic and a second opinion, never the headline predictor. The as-driven estimator with the learned factor predicts the race better. The diagnostic's value is explanatory: it tells you why a Friday shows no signal and it explains about two-thirds of the pooled Friday-to-Sunday gap on the 16 issued cases (68.5% pooled; per-case shares vary widely).

4. Why not a Bayesian hierarchical model, as your idea submission promised?

Because with eleven weekends the honest first step is the simplest estimator that validates, and it validates. The hierarchical layer is exactly what the three-season download is for: partial pooling of the transfer factor by compound across seasons, with a coverage check on the bands. It is on the roadmap for after the event, not because it is hard, but because it should be built on more than one season.

5. Where could information leak from the race into the Friday prediction?

Three places, all closed. The transfer factor for a weekend is computed from other weekends only. The fallback value is the median of other weekends' withheld outcomes. Compound offsets come from that weekend's practice or qualifying medians (a nominal Pirelli-range step where the measured step is implausible), all of which precede the race. The only thing that sees the race is the scorecard.

6. Would this work in Formula 2, where there is no telemetry feed?

The estimator needs lap times, tyre age and compound, and flags, which F2 timing provides; fuel and evolution corrections carry over. What is lost without telemetry is the traffic flag and the push proxy, so more laps would be withheld and the bands would widen. That is the honest degraded mode, and it is still far better than a straight line.

7. What happens when Pirelli brings different compounds to a circuit next year?

The curve is per compound name at that weekend, learned from that weekend's Friday; the transfer factor is per compound role, soft, medium, hard, which is about how drivers treat the role. If the role changes, the agreement rule notices the ratios disagree and stops applying the factor until enough weekends agree again.

8. Half your withheld cases are hard tyres with too few Friday laps. Is that a method problem or a data problem?

A data problem that the method reports rather than hides: teams rarely run the hard on Friday at European rounds. The fallback still gives a forecast, and the practice-to-sprint test on sprint weekends adds a second observation of the hard. A team with its own Friday plan could fix it in an afternoon by running the hard.

9. If you had one more day, what would you build first?

The per-driver degradation and push profile, teammate against teammate, checked for repeatability across weekends. It answers a question strategists and broadcasters actually ask, who looks after their tyres, and it needs nothing we do not already have.

10. You say live monitoring, but the race starts after the event closes. What are you actually showing?

A lap-by-lap replay of a race we have already scored, Monza or Hungary: the Friday forecast on screen, then the live estimator updating at lap 10, 20 and 30, the band shrinking, the alerts firing or not, and the pit call moving or holding. The same code runs on Madrid on Sunday evening from the live-timing stream; the forecast it will be compared against is hashed before the race (sha256 8e5489d5824a, published 21:33 IST).